

# Python Assignment

You've just joined a small but ambitious startup. The team moves fast and is spread across multiple time zones. Ideas are exchanged in online meetings.

This resulting specs and tasks end up scattered everywhere: sticky notes, chat messages, personal to-do lists.

Right now, it's pure chaos.

During a stand-up, the Product Owner says:

"We really need a simple task service. Something internal. Clean. We'll build more on top of it later."

Everyone agrees it's a great idea. You volunteer to take ownership of the project, and after aligning with the team, it becomes clear that this service should meet the following **minimum requirements**:

1. Task Entity with the following properties:
  - A unique numeric ID
  - A title (which must be unique — no confusion in stand-ups)
  - An optional description
  - A status (new, in progress and completed)
  - A priority between 1 (low) and 5 (high)
  - A creation timestamp
2. Use in-memory repository:
  - Provide CRUD operations
  - Raise exceptions when errors occur
3. Based on FASTAPI which supports:
  - Creating tasks
  - Listing all tasks
  - Fetching a single task
  - Updating a task
  - Deleting a task

The team also asks for small but useful things:

- Filter tasks by status
  - Sort tasks by priority
4. Testing
    - Do your best to include tests
  5. Project Structure
    - Provide a README explaining the project structure
    - Explain how to run the application and tests.